

GRUNDLAGEN ZU ABLEITUNGEN UND INTEGRALEN

* **Ableitungen.** Geben Sie für die folgenden Funktionen den Definitionsbereich und die Ableitung an:

1. $f(x) = \sqrt[4]{x^3}$ Wegen $\sqrt[n]{\dots}: x^3 \geq 0$ für $x \geq 0$: $\mathbb{D} = [0, \infty[= \mathbb{R}_0^+$

$$f'(x) = \left(x^{\frac{3}{4}}\right)' = \frac{3}{4} x^{\frac{3}{4}-1} = \frac{3}{4} x^{-\frac{1}{4}} = \frac{3}{4} \frac{1}{\sqrt[4]{x}}$$

2. $f(x) = \sqrt[3]{\frac{x^2}{x}}$ $\mathbb{D} = \mathbb{R} \setminus \{0\}$ ($f(x) = x^{2-\frac{1}{3}} = x^{\frac{5}{3}} = \sqrt[3]{x^5}$ $\mathbb{D} = \mathbb{R}$)

$$f'(x) = \left(x^{\frac{5}{3}}\right)' = \frac{5}{3} x^{\frac{5}{3}-1} = \frac{5}{3} x^{\frac{2}{3}} = \frac{5}{3} \sqrt[3]{x^2}$$

3. $f(x) = -10x^4 + 2x^3 - 2$ Polynom $\mathbb{D} = \mathbb{R}$

$$f'(x) = -40x^3 + 6x^2$$

4. $h(x) = a \cos(x) - x^2 + e^x + 1$, $\mathbb{D} = \mathbb{R}$
 ↑ konstante/feste Zahl

$$h'(x) = a(-\sin x) - 2x + e^x = -a \sin x - 2x + e^x$$

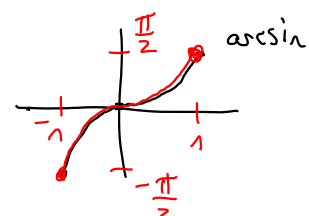
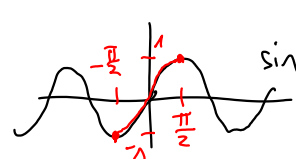
5. $f(x) = 2x \ln x$ $\mathbb{D} = \mathbb{R}^+ =]0, \infty[$
 Produktregel

$$f'(x) = 2 \cdot \ln x + 2x \cdot \frac{1}{x} = 2 \ln x + 2$$

6. $f(x) = x^2 \arcsin(x)$ $\mathbb{D} = [-1, 1]$

Produktregel

$$f'(x) = 2x \cdot \arcsin(x) + x^2 \cdot \underbrace{\left(\arcsin(x)\right)'}_{\frac{1}{\sqrt{1-x^2}} \text{ (Vorlesung)}}$$



7. $f(x) = \sin x \cos x$ $D = \mathbb{R}$
 Produktregel

$$f'(x) = \cos x \cos x + \sin x (-\sin x) = \cos^2 x - \sin^2 x \stackrel{\sin^2 x + \cos^2 x = 1}{=} \cos^2 x - (1 - \cos^2 x) = 2\cos^2 x - 1$$

8. $f(x) = \frac{5x^2 - 6x + 1}{x^2 + 2x + 1}$ NST Nenner: $x^2 + 2x + 1 = (x+1)^2$ NST $x_{1,2} = -1$, $D = \mathbb{R} \setminus \{-1\}$

Quotientenregel

$$f'(x) = \frac{(x^2 + 2x + 1)' \cdot (5x^2 - 6x + 1) - (5x^2 - 6x + 1)' \cdot (x^2 + 2x + 1)}{(x^2 + 2x + 1)^2} = \frac{(2x+2) \cdot (5x^2 - 6x + 1) - (10x - 6) \cdot (x+1)^2}{(x+1)^4} = \frac{16x - 8}{(x+1)^3}$$

9. $f(x) = \arccos(\sqrt{x^2 - 1})$: $-1 \leq \sqrt{x^2 - 1} \leq 1 \Leftrightarrow \sqrt{x^2 - 1} \leq 1 \stackrel{(\cdot)^2}{\Leftrightarrow} x^2 - 1 \leq 1 \Leftrightarrow x^2 \leq 2$
 erfüllt, da ≥ 0 $\sqrt{x^2 - 1} \geq -1$ str. wo. wa. auf $[0, \infty]$

$x^2 - 1 \geq 0 \Leftrightarrow x^2 \geq 1$ Insgesamt: $D = [-\sqrt{2}, -1] \cup [1, \sqrt{2}]$

$$f'(x) = (\arccos(\sqrt{x^2 - 1}))' = -\frac{1}{\sqrt{1 - (\sqrt{x^2 - 1})^2}} \cdot (\sqrt{x^2 - 1})' = -\frac{1}{\sqrt{1 - x^2 + 1}} \cdot \frac{1}{2\sqrt{x^2 - 1}} \cdot (x^2 - 1)' = -\frac{x}{\sqrt{2 - x^2} \sqrt{x^2 - 1}}$$

10. $f(x) = x^x = e^{x \ln x}$ für $x > 0$, d.h. $D = \mathbb{R}^+$ $(0^0 \text{ Video Durchsch. bei negativen Werten nur manchmal: } (-\frac{1}{2})^{-\frac{1}{2}} = \frac{1}{\sqrt{-1/2}} \checkmark \text{ vs. } (-\frac{1}{2})^{-\frac{1}{2}} = \frac{1}{\sqrt{-1/2}} \text{ ✗})$

$$f'(x) = (e^{x \ln x})' = e^{x \ln x} \cdot (x \ln x)' = x^x (\ln x + 1)$$

Bem.: Man kann f stetig auf $\mathbb{R}_0^+$ fortsetzen mit $\lim_{x \rightarrow 0} e^{x \ln x} = e^{\lim_{x \rightarrow 0} x \ln x} = e^0 = 1$, $\lim_{x \rightarrow 0} \frac{\ln x}{\frac{1}{x}} \stackrel{e'H}{=} \lim_{x \rightarrow 0} \frac{-\frac{1}{x}}{-\frac{1}{x^2}} = \lim_{x \rightarrow 0} -x = 0$

11. $f(x) = (\frac{x}{1+x})^n$ ($n \in \mathbb{N}$)

$D = \mathbb{R} \setminus \{-1\}$, $f'(x) = \left(\left(\frac{x}{1+x} \right)^n \right)' \stackrel{\text{Kettenregel}}{=} n \left(\frac{x}{1+x} \right)^{n-1} \cdot \left(\frac{x}{1+x} \right)' = n \left(\frac{x}{1+x} \right)^{n-1} \cdot \frac{1}{(1+x)^2} = \frac{n x^{n-1}}{(1+x)^{n+1}}$

12. $f(x) = 2 \ln(x^3 - 2x)$: $x^3 - 2x > 0 \Leftrightarrow x(x^2 - 2) > 0$ NST $0, \pm\sqrt{2}$: $D =]-\sqrt{2}, 0[\cup]\sqrt{2}, \infty[$

Kettenregel

$$f'(x) = 2 \cdot \frac{1}{x^3 - 2x} \cdot (x^3 - 2x)' = 2 \cdot \frac{3x^2 - 2}{x^3 - 2x}$$

13. $f(x) = 5e^{-x^2}$: $D = \mathbb{R}$

Kettenregel

$$f'(x) = 5 \cdot e^{-x^2} \cdot (-x^2)' = -10x \cdot e^{-x^2}$$

14. $x(t) = \ln(\frac{1}{t^2}) + \ln(\frac{t+4}{t})$: $\frac{1}{t^2} > 0$ $\frac{t+4}{t} > 0 \Leftrightarrow \begin{cases} t > 0 \\ \Leftrightarrow t+4 > 0 \Leftrightarrow t > -4: t > 0 \end{cases}$
 $\Leftrightarrow \begin{cases} t < 0 \\ \Leftrightarrow t+4 < 0 \Leftrightarrow t < -4: t < -4 \end{cases}$
 $D =]-\infty, -4[\cup]0, \infty[$

$$x'(t) = \frac{1}{\frac{1}{t^2}} \cdot \left(\frac{1}{t^2} \right)' + \frac{1}{\frac{t+4}{t}} \cdot \left(\frac{t+4}{t} \right)' = -2 \frac{t^2}{t^3} + \frac{t}{t+4} \cdot \frac{-4}{t^2} = -\frac{2}{t} - \frac{4}{(t+4) \cdot t}$$

$$\text{ODER: } \ln\left(\frac{1}{t^2}\right) = \ln(t^{-2}) = -2 \ln t \xrightarrow{\text{ableiten}} -\frac{2}{t}$$

$$\text{ODER: } \ln\left(\frac{1}{t^2}\right) = \underbrace{\ln(1)}_0 - \ln(t^2) \xrightarrow{\text{ableiten}} -\frac{2}{t}$$

↖ Kettenregel

Eigener Lösungsversuch.

1. $f(x) = \sqrt[4]{x^3}$

2. $f(x) = \frac{x^2}{\sqrt[3]{x}}$

3. $f(x) = -10x^4 + 2x^3 - 2$

4. $h(x) = a \cos(x) - x^2 + e^x + 1$

5. $f(x) = 2x \ln x$

6. $f(x) = x^2 \arcsin(x)$

7. $f(x) = \sin x \cos x$

8. $f(x) = \frac{5x^2 - 6x + 1}{x^2 + 2x + 1}$

9. $f(x) = \arccos(\sqrt{x^2 - 1})$

10. $f(x) = x^x$

11. $f(x) = \left(\frac{x}{1+x}\right)^n \quad (n \in \mathbb{N})$

12. $f(x) = 2 \ln(x^3 - 2x)$

13. $f(x) = 5e^{-x^2}$

14. $x(t) = \ln\left(\frac{1}{t^2}\right) + \ln\left(\frac{t+4}{t}\right)$

Zweite Ableitungen. Berechnen Sie die zweiten Ableitungen:

1. $f(x) = x \sin x$

$$f'(x) = 1 \cdot \sin x + x \cdot \cos x$$

$$f''(x) = \cos x + 1 \cdot \cos x + x \cdot (-\sin x) = 2 \cos x - x \cdot \sin x.$$

2. $f(x) = x^2 + \ln x$

$$f'(x) = 2x + \frac{1}{x}$$

$$f''(x) = 2 - \frac{1}{x^2}$$

$$\left(\frac{1}{x}\right)' = (x^{-1})' = -x^{-2} = -\frac{1}{x^2}.$$

3. $f(x) = \cos(x^2)$

$$f'(x) = -\sin(x^2) \cdot 2x = -2x \cdot \sin(x^2)$$

$$f''(x) = -2 \sin(x^2) + (-2x) \cdot \underbrace{(\sin(x^2))'}_{\cos(x^2) \cdot 2x} = -2 \sin(x^2) - 4x^2 \cos(x^2).$$

Eigener Lösungsversuch.

1. $f(x) = x \sin x$

2. $f(x) = x^2 + \ln x$

3. $f(x) = \cos(x^2)$

Polynom dritten Grades. Gesucht ist ein Polynom $p(x)$ mit

$$\underline{p(3) = 4}, \quad \underline{p'(3) = -2}, \quad \underline{p''(3) = 7}, \quad \underline{p'''(3) = -3}.$$

Lösung.

Ausatz: $p(x) = ax^3 + bx^2 + cx + d$

$$p'(x) = 3ax^2 + 2bx + c$$

$$p''(x) = 6ax + 2b$$

$$p'''(x) = 6a$$

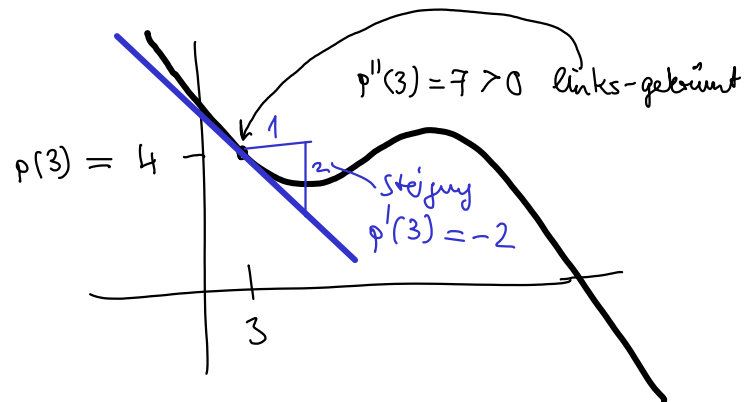
$$\underline{p(3) = 27a + 9b + 3c + d \stackrel{!}{=} 4} \Rightarrow d = 55$$

$$\underline{p'(3) = 27a + 6b + c \stackrel{!}{=} -2} \Rightarrow c = -\frac{73}{2}$$

$$\underline{p''(3) = 18a + 2b \stackrel{!}{=} 7} \Rightarrow b = 8$$

$$\underline{p'''(3) = 6a \stackrel{!}{=} -3} \Rightarrow a = -\frac{1}{2}$$

$$\Rightarrow p(x) = -\frac{1}{2}x^3 + 8x^2 - \frac{73}{2}x + 55$$



Eigener Lösungsversuch.

* **Integrale.** Berechnen Sie:

Unbestimmte Integrale

1. $\int 3 \cos(x) dx = 3 \sin(x) + C$ mit $C \in \mathbb{R}$ ← Menge aller Stammfkten = **Unbestimmtes Integral**

⚠ nicht vergessen!

$$\int x^n dx = \begin{cases} \frac{x^{n+1}}{n+1} & n \neq -1 \\ \ln|x| & n = -1 \end{cases} \quad n \in \mathbb{R}!$$

2. $\int \frac{2}{x^2} dx = 2 \int x^{-2} dx = 2 \cdot \frac{x^{-2+1}}{-2+1} + C = 2 \cdot \frac{x^{-1}}{-1} + C = -2 \cdot \frac{1}{x} + C$

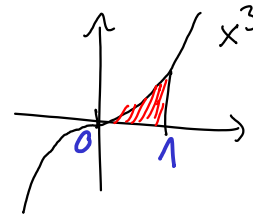
3. $\int 2x^3 + 4x dx = 2 \frac{x^4}{4} + 4 \frac{x^2}{2} + C = \frac{1}{2} x^4 + 2x^2 + C$

4. $\int e^t - 3t^2 dt = e^t - 3 \frac{t^3}{3} + C = e^t - t^3 + C$
 $\uparrow$
 $\int e^t dt = e^t + C$

5. $\int \frac{x^2 \cdot \sqrt{x}}{\sqrt[3]{5}} dx = \frac{1}{\sqrt[3]{5}} \cdot \int x^{2+\frac{1}{2}} dx = \frac{1}{\sqrt[3]{5}} \cdot \frac{x^{\frac{5}{2}+1}}{\frac{5}{2}+1} + C = \frac{1}{\sqrt[3]{5}} \cdot \frac{x^{\frac{7}{2}}}{\frac{7}{2}} + C = \frac{2}{7 \cdot \sqrt[3]{5}} \cdot \sqrt{x^7} + C$

bestimmte Integrale
= Fläche unter dem Graphen

6. $\int_0^1 x^3 dx = \left[\frac{x^4}{4} \right]_0^1 = \frac{1^4}{4} - \frac{0^4}{4} = \frac{1}{4}$



7. $\int_0^1 (e^t + 2t) dt = [e^t + t^2]_0^1 = (e^1 + 1^2) - (\underbrace{e^0 + 0^2}_1) = e \approx 2,71...$

8. $\int_0^2 \frac{x^2}{1+x^3} dx = \frac{1}{3} \int_0^2 \frac{1}{1+x^3} \cdot 3x^2 dx = \frac{1}{3} \cdot \left[\ln|1+x^3| \right]_0^2 = \frac{1}{3} \left(\underbrace{\ln|1+2^3|}_9 - \underbrace{\ln|1+0^3|}_0 \right) = \frac{1}{3} \ln 9$
 (Kettenregel) (Substitutionsregel (zspäter))

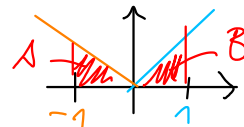
Hier Spezialfall der Substitutionsregel: **Logarithmische Integration**

$$\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C$$

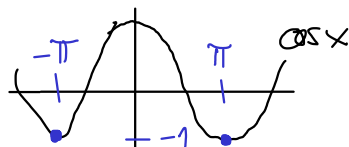
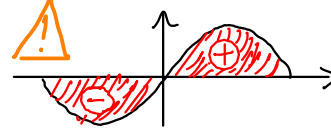
$$9. \int_0^1 x^2 + 1 dx = \left[\frac{x^3}{3} + x \right]_0^1 = \left(\frac{1^3}{3} + 1 \right) - \underbrace{\left(\frac{0^3}{3} + 0 \right)}_0 = \frac{4}{3}$$

$$10. \int_{-1}^1 |x| dx = \int_{-1}^0 \underbrace{-x}_{\text{A}} dx + \int_0^1 \underbrace{x}_{\text{B}} dx = \left[-\frac{x^2}{2} \right]_{-1}^0 + \left[\frac{x^2}{2} \right]_0^1 = \left(0 + \frac{1}{2} \right) + \left(\frac{1}{2} - 0 \right) = 1$$

⚠ Vorsicht bei abschnittsweise definierten Funktionen: $|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$



$$11. \int_{-\pi}^{\pi} \sin(x) dx = \left[-\cos(x) \right]_{-\pi}^{\pi} = \underbrace{-\cos(\pi)}_{-1} + \underbrace{\cos(-\pi)}_{-1} = 0$$



Eigener Lösungsversuch.

1. $\int 3 \cos(x) dx$

2. $\int \frac{2}{x^2} dx$

3. $\int 2x^3 + 4x dx$

4. $\int e^t - 3t^2 dt$

5. $\int \frac{x^2 \cdot \sqrt{x}}{\sqrt[3]{5}} dx$

6. $\int_0^1 x^3 dx$

7. $\int_0^1 (e^t + 2t) dt$

8. $\int_0^2 \frac{x^2}{1+x^3} dx$

9. $\int_0^1 x^2 + 1 dx$

10. $\int_{-1}^1 |x| dx$

11. $\int_{-\pi}^{\pi} \sin(x) dx$